



Context

Traffic congestion causes heavy delays, significant economic losses and severe pollution — more than 250 billion CFA francs per year in Senegal.

The BRT improves mobility, but creates new challenges at intersections, where coordination between buses and conventional traffic becomes complex.



Issue

Current traffic light and intersection management systems are static, inefficient, and unable to adapt to real-time traffic. Note:

- Recurring congestion on the Dakar BRT corridor.
- Difficulty in maintaining traffic flow while respecting BRT priority.
- Limitations of fixed lights and manual regulation.

Goals

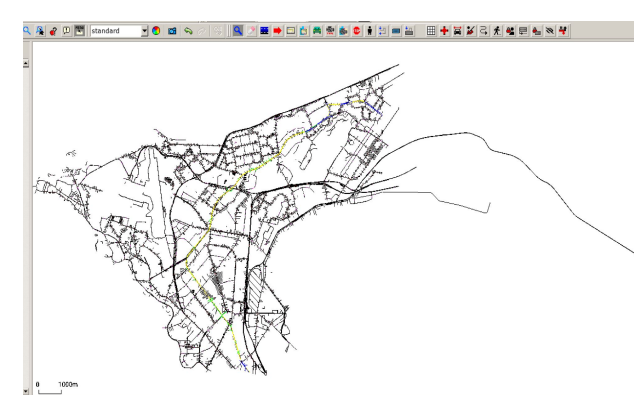
Improve traffic flow on the BRT corridor while respecting its priority.

- Reduce average waiting times at intersections.
- Minimize queues at intersections
- Reduce journey delays.
- Reduce average time lost per vehicle

Methodology



Actual map of Dakar

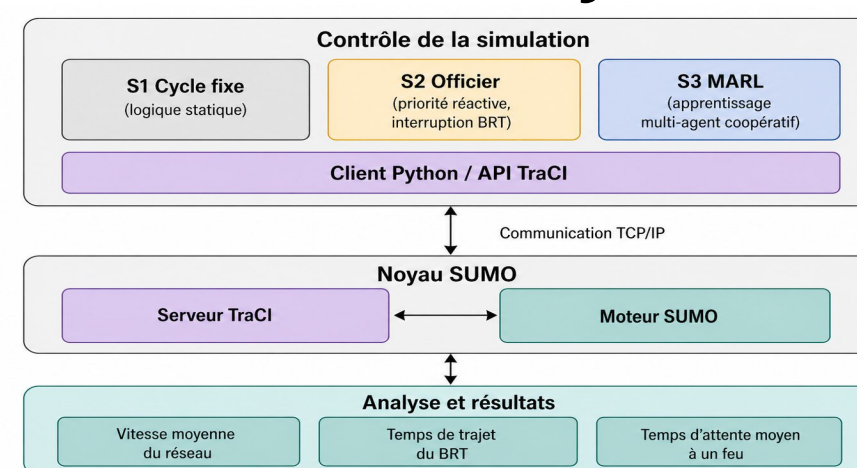


Transformation into an environment virtual for SUMO



Training AI agents in SUMO (MARL)

Overview of how the system works



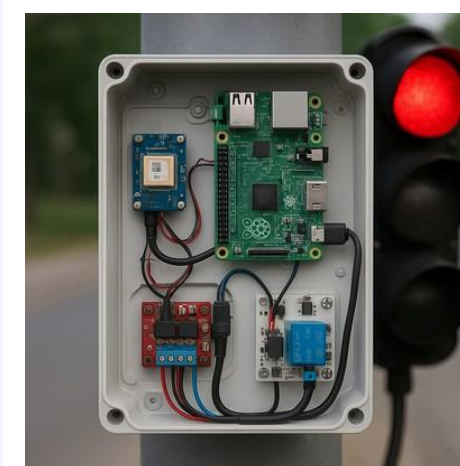
System characteristics

- 150 simulated intersections.
- 2020 vehicles.
- 23 BRT stations.
- Decentralized learning (DTDE).
- Fitted Q-Iteration + Random Forest.
- Local coordination between intersections.

Expected impact

Optimizing intersection management can lead to:

- Reduced time wasted at intersections, improved air quality in the neighborhood, fewer agents needed, and better integration with the BRT.
- Improved management and priority routes for emergency vehicles



Perspectives

- Pedestrian integration. Explicit
- communication between agents. Testing
- under heavy traffic. Deployment on a real corridor. Integration of data from real sensors.

References

- Large-scale Traffic Signal Control Using a Novel Multi-Agent Reinforcement Learning (<https://arxiv.org/pdf/1908.03761>)
- SUMO-RL (<https://github.com/LucasAlegre/sumo-rl>)
- SUMO (<https://sumo.dlr.de/docs/index.html>)

